

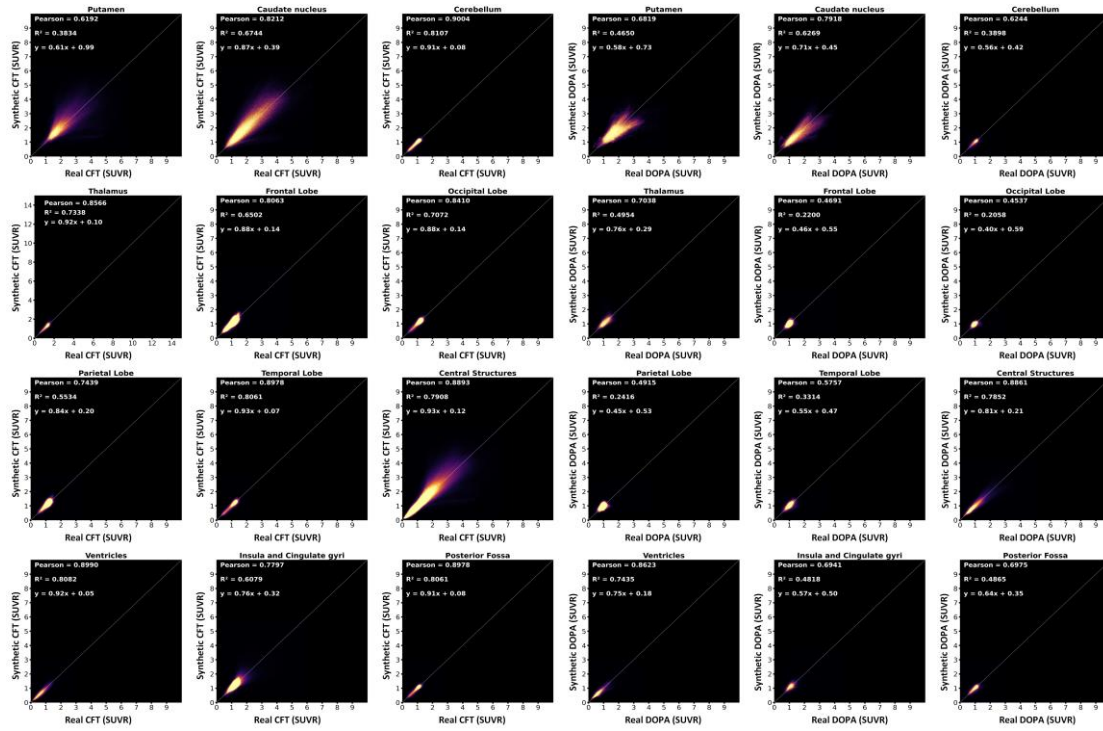


Figure S1. SUVR-based region-wise joint histograms and linear regression analyses of the 12 regions of interest based on the real and synthetic ¹¹C-CFT/¹⁸F-DOPA PET images.

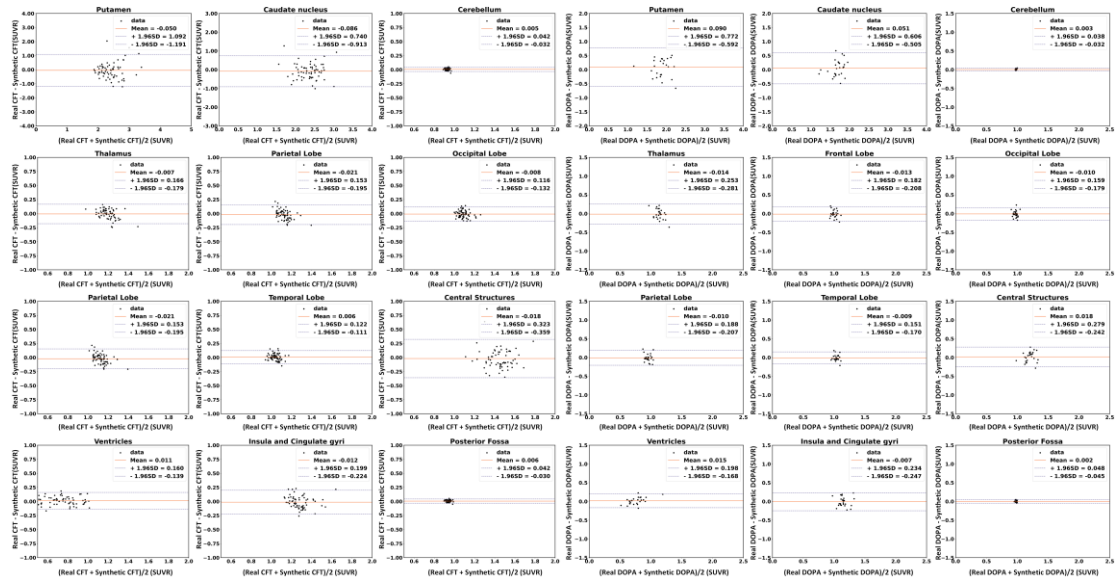


Figure S2. Bland-Altman analysis of SUVR values between the real and synthetic ^{11}C -CFT/ ^{18}F -DOPA PET images across 12 regions of interest.

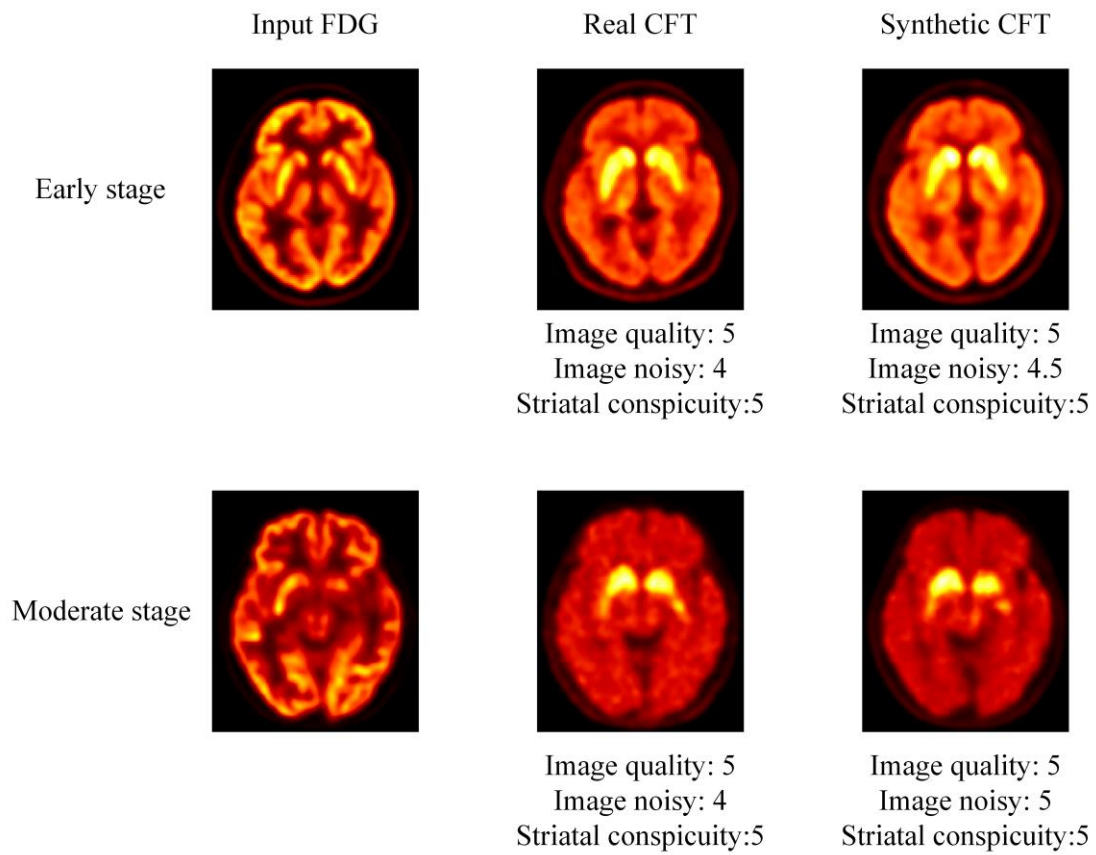


Figure S3. Representative ^{18}F -FDG (input), real ^{11}C -CFT, and synthetic ^{11}C -CFT images from an early-stage (top) and a moderate-stage (bottom) PD patient, selected by an experienced nuclear medicine physician from the test set. The scores shown beneath each image are derived from the clinical evaluation, where two nuclear medicine physicians independently rated anonymized images.

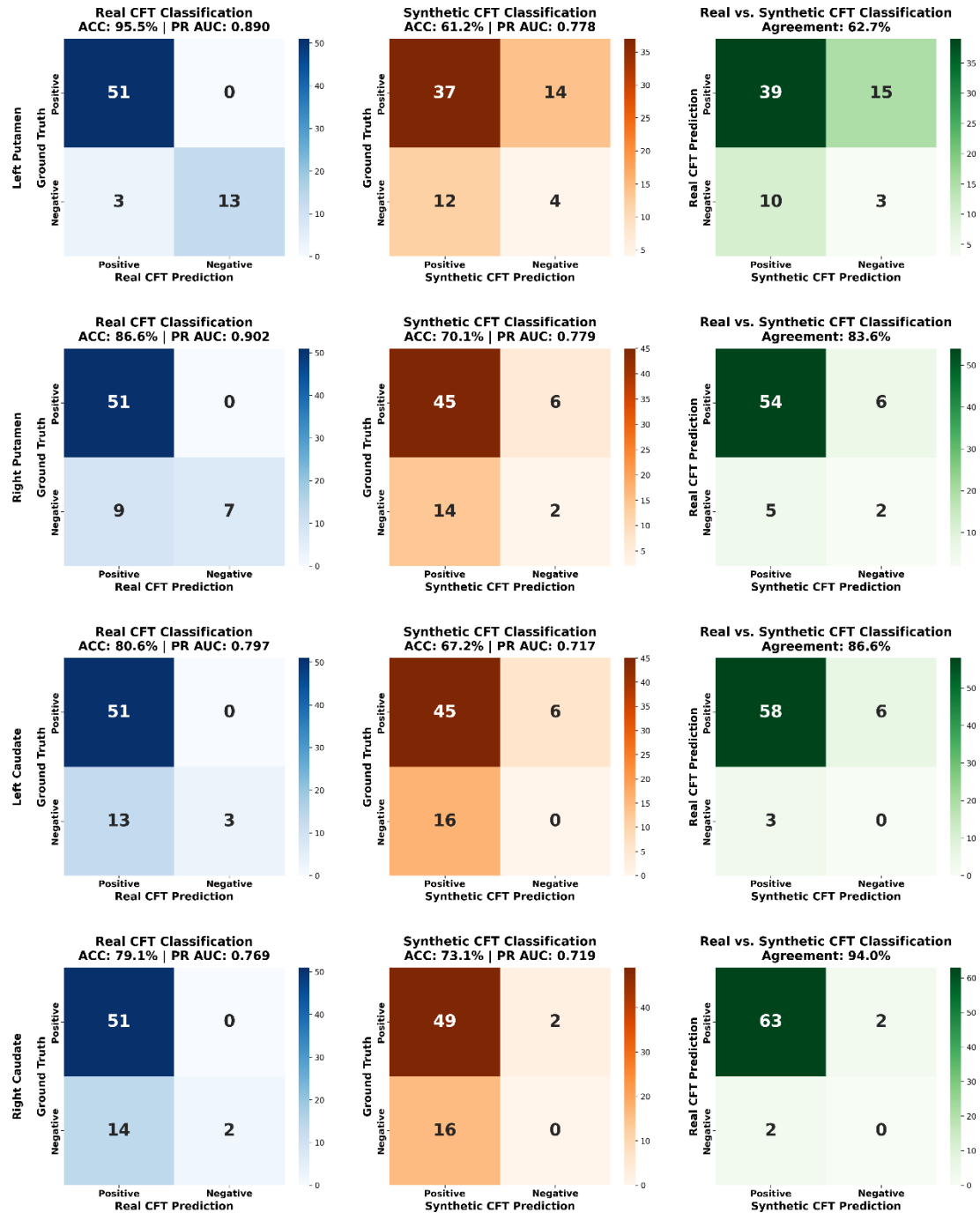


Figure S4. Comprehensive diagnostic agreement analysis across four striatal regions for the ^{11}C -CFT cohort. Columns from left to right illustrate the diagnostic accuracy of real PET versus the clinical ground truth, the accuracy of synthetic PET versus the ground truth, and the direct diagnostic agreement between real and synthetic predictions.

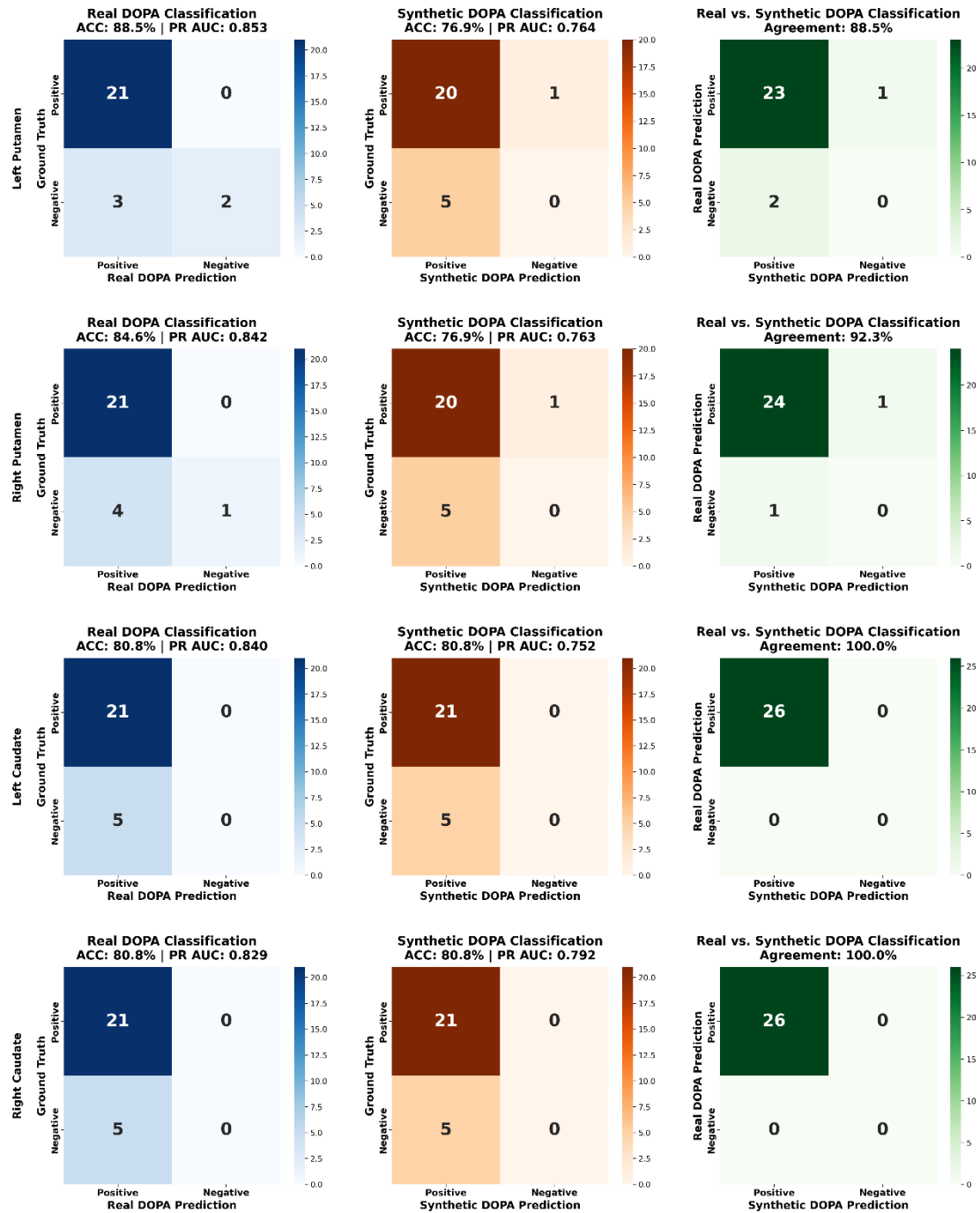


Figure S5. Comprehensive diagnostic agreement analysis across four striatal regions for the ^{18}F -DOPA cohort. Columns from left to right illustrate the diagnostic accuracy of real PET versus the clinical ground truth, the accuracy of synthetic PET versus the ground truth, and the direct diagnostic agreement between real and synthetic predictions.